

Yijun LIU

Sydney, Australia · yijunliu931@gmail.com · <https://yijunliu931.github.io>
0452586688

Summary

- Backend engineer with 5+ years of experience in Go, Python, and Java. Designed and built two production platforms from scratch at CertiK — Terraflow (15–20M tx/day data pipeline) and SkyNode (fullstack validator infrastructure). Track record of delivering cost savings: 70–80% via cloud migration, ~\$50K/year via Python-to-Go rewrite.
- Built nsa-auditor, a neuro-symbolic AI security auditor using LLM agent pipelines and symbolic verification. Practitioner of Harness Engineering — AI-assisted development with automated quality gates, compressing multi-month projects to weeks.

Technical Skills

Programming Languages	Python, Golang, Java, C++
Frontend	React (CertiK Staking Hub fullstack)
Cloud	AWS, Azure (Data Factory, Blob Storage, Azure SQL), Hetzner, OVH, Latitude
DevOps & Infrastructure	Terraform, Docker, Kubernetes, GitHub Actions, CI/CD, Prometheus, Grafana
Database	PostgreSQL, MySQL, MongoDB, TiDB, Redis

Work Experience

CertiK

Software Engineer III

Remote

Feb. 2024 - Present

- Designed and built Terraflow from scratch — a real-time Python data ingestion platform processing 15–20 million transactions per day, including architecture design, database schema, pluggable adapter pattern for 10+ data sources, RESTful/gRPC APIs, and automated data validation pipelines.
- Built the entire SkyNode infrastructure from the ground up — provisioned and configured production Kubernetes clusters, Docker containerisation, Prometheus/Grafana monitoring and alerting, and automated deployment pipelines across multiple cloud providers.
- Developed the CertiK Staking Hub (skynode.certik.com) fullstack — React frontend with staking dashboard, user management, and real-time validator monitoring; Go backend with API services and database integration for persistent data storage.
- Scaled SkyNode from a handful of deployments to 15+ production validator nodes across multiple networks; provided technical guidance and on-call mentoring to team members across multiple time zones.
- Leading the migration of Terraflow from Python to Go with byte-for-byte output parity, using Harness Engineering methodology — coding agents as primary producers with a record/replay parity harness, targeting ~\$50K/year infrastructure savings.
- Migrated infrastructure from AWS to cost-efficient providers (Hetzner, OVH, Latitude), reducing operational costs by 70–80% while maintaining production reliability and SLA compliance.

Varis (ODP Corporation)

Software Developer II

USA

Feb. 2021 - Jan. 2024

- Drove the technical migration from Java (Spring Boot) to Golang, redesigning core microservices (Gin, gRPC, SQL/NoSQL) for a high-concurrency B2B e-commerce platform, improving performance and maintainability.
- Designed and developed scalable backend APIs and business logic layers integrating with PostgreSQL and Azure SQL, handling concurrent requests from multiple client applications and third-party systems.
- Engineered distributed processing solutions to handle large-scale supplier catalog ingestion (2–30GB files) using streaming, Kafka event-driven architecture, and Azure Data Factory for ETL orchestration.
- Led the migration from AWS to Azure (Data Factory, Blob Storage, Azure SQL), redesigning data pipelines and reducing infrastructure costs while maintaining service continuity and performance.
- Implemented CI/CD pipelines and automated deployment workflows; built operational dashboards and health checks to monitor production services across Azure environments.
- Drove engineering excellence through unit testing (JUnit, Go testing), code reviews, and microservices best practices; mentored junior engineers on Go adoption, system design, and production debugging workflows.

AI & Agentic Projects

- Built nsa-auditor — a neuro-symbolic smart-contract security auditor: LLM agent pipeline proposes vulnerabilities, symbolic verifier (Halmos/Z3) mathematically proves them. 4-stage orchestration (planner → spec author → verifier → adjudicator), RAG comparison leg with Qdrant vector DB, and evaluation harness with recall/precision metrics.
- Harness Engineering Practice: agent-ready repository for Python→Go migration with record/replay parity harness, mechanically-enforced architecture rules, cross-session memory, and weekly entropy-reduction workflows.

Education

University of Wyoming

Master of Science in Computer Science (GPA 3.89/4.0)

Wyoming, USA

Sep. 2018 - Aug. 2020

Dalian/Wuhan University of Technology

Master/Bachelor of Engineering

Dalian/Wuhan, China

Sep. 2010 - July 2016